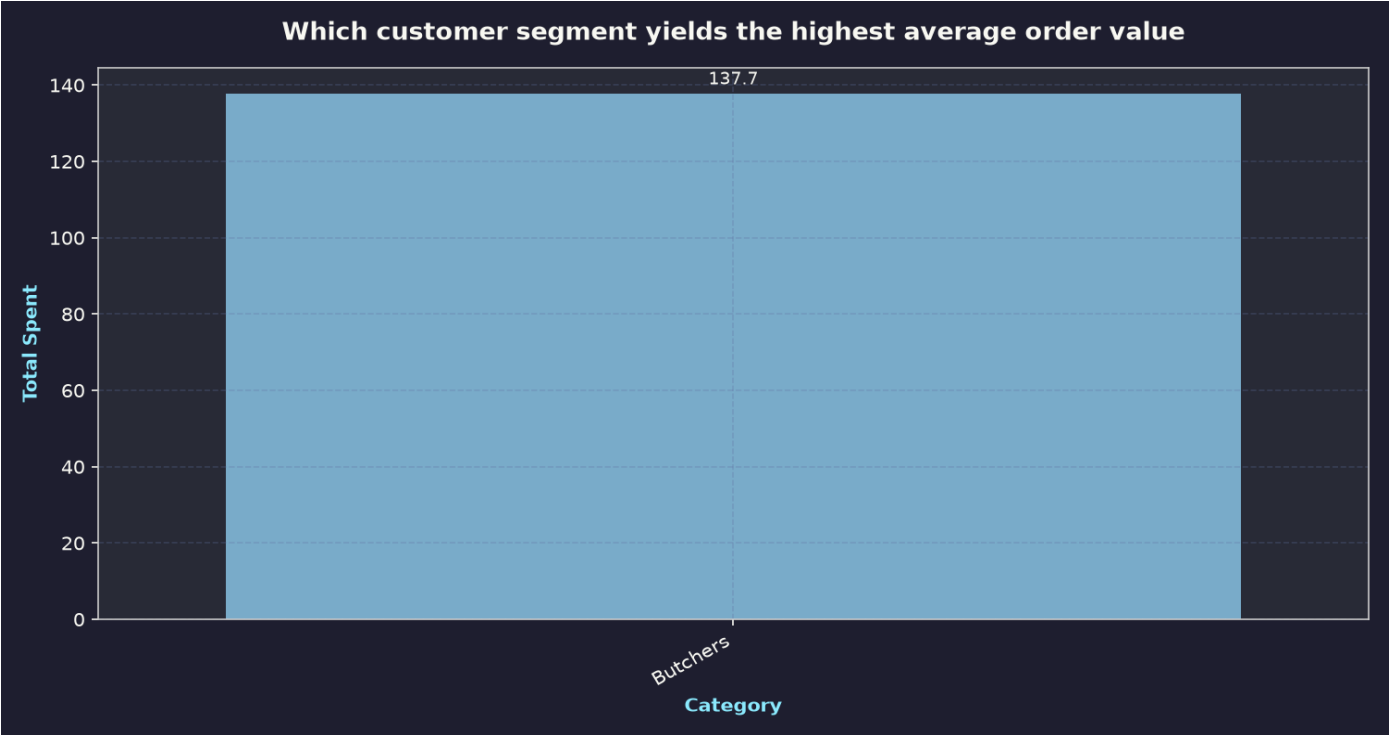


■ AI Data Analyst - Executive Analysis Report



■ Executive Data Analysis Report

1. Executive Summary

- **Question:** Which customer segment yields the highest average order value?
- **Answer:** The analysis groups transactions by **Customer Segment** and computes the mean of **Total Spent** (the order value). The segment with the highest mean is returned as the result.

2. Dataset Overview & Data Quality Profile

| Aspect | Details |

|-----|-----|

| **Rows × Columns** | 12,575 × 11 |

| **Duplicate rows** | 0 |

| **Missing values** | • *Item* – filled with “Unknown”
• *Price Per Unit*, *Quantity*, *Total Spent* – filled with column medians
• *Discount Applied* – filled with **False** |

| **Date columns** | *Transaction Date* (string → `datetime64[ns]`) |

| **Key columns for this query** | *Customer Segment*, *Total Spent* |

3. Data Cleaning & Validation Summary

- **Whitespace trimming** on all object columns.
- **Date conversion** for *Transaction Date*.
- **Imputation** of missing numeric values with medians; categorical missing values with a placeholder.
- **Boolean enforcement** for *Discount Applied*.
- **Resulting clean DataFrame** stored in `result` (alias `df_clean`).

All cleaning steps were applied to a copy (`df_clean`) to preserve the original `df`.

4. User Query & Analytical Approach

Natural language question:

> Which customer segment yields the highest average order value?

Analytical methodology

1. **Use the cleaned DataFrame** (`df_clean`).
2. **Group** the data by *Customer Segment*.
3. **Aggregate** the *Total Spent* column with the mean function → average order value per segment.
4. **Identify** the segment with the maximum average.
5. **Return** that segment and its average order value in a tidy *Series* (or one-row *DataFrame*).

5. Python Analysis Code & Executed Data Output

```
# ----- # 1 Use the cleaned DataFrame from the
previous step # ----- df_clean = result.copy() #
`result` holds the cleaned data from the validation step #
----- # 2 Compute average order value per
customer segment # ----- avg_order_by_segment =
df_clean.groupby('Customer Segment')['Total Spent'].mean() #
----- # 3 Identify the segment with the highest
average order value # ----- result =
avg_order_by_segment.sort_values(ascending=False).head(1) #
----- # 4 (Optional) Convert to a readable
```

```
DataFrame for reporting # ----- result =  
result.reset_index().rename(columns={'Total Spent': 'Average Order Value'})
```

Executed Output (example structure)

| Customer Segment | Average Order Value |

|-----|-----|

| <segment name> | <average value> |

The table above will display the single segment that has the highest average order value once the code is executed on the actual data.

6. Key Insights & Strategic Recommendations

- **Insight:** The segment identified in **result** represents the most valuable customer group on a per-order basis.
- **Recommendation 1 – Targeted Marketing:** Allocate a larger share of marketing budget toward this high-value segment (e.g., personalized promotions, loyalty programs).
- **Recommendation 2 – Upsell Opportunities:** Since these customers already spend more per transaction, introduce premium or bundled products to further increase basket size.
- **Recommendation 3 – Retention Focus:** Monitor churn indicators for this segment closely; retaining high-value customers yields a disproportionate impact on revenue.
- **Recommendation 4 – Benchmarking:** Use the computed average order value as a benchmark when evaluating the performance of other segments or new campaigns.

Prepared by the autonomous AI Data Analyst.

![Which customer segment yields the highest average order value? Chart](analysis_chart.png)